

In-Class Activity: “Clinical NLP in 45 Minutes”

Learning Objectives

By the end of this activity, you should be able to:

- Use Hugging Face pipelines for real inference.
 - Understand model-task alignment (classification vs. generation).
 - Critically evaluate outputs in a health context (bias, hallucination, risk).
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Deliverables (End of Class)

Each team must submit work in a shared repository.

- The repository must contain the code used for the activity.
 - Each team member must have at least one commit associated with their name.
 - Your written answers for each section can be added in either a Jupyter notebook or a Markdown file.
 - We will discuss results together at the end of class.
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Setup

Work locally in Python from the terminal. Create an environment and install the required packages with uv:

```
uv venv
source .venv/bin/activate
uv pip install transformers datasets torch
```

Then create a Python file such as `activity.py` and run it with:

```
python activity.py
```

Part 1: Clinical Text Classification

Your Task

In this part, everyone should use the same pretrained model so the class can compare results:

```
distilbert/distilbert-base-uncased-finetuned-sst-2-english
```

This is **not** a medical model. It is a general English sentiment classifier trained on SST-2. You will use it on clinical notes on purpose to see what goes wrong when the model and task do not match.

Run the code below exactly as written.

Minimal Code

```
from transformers import pipeline

classifier = pipeline(
    "text-classification",
    model="distilbert/distilbert-base-uncased-finetuned-sst-2-english"
)
```

```

texts = [
    "Patient reports chest pain and shortness of breath.",
    "Routine follow-up visit for blood pressure check.",
    "Mild headache for two days, no other symptoms."
]

for t in texts:
    print(t, "->", classifier(t))

```

Answer

After you run the model, answer these questions in 2–3 sentences each:

- What labels does the model return?
 - Are those labels a good fit for clinical triage? Why or why not?
 - Based on the model card, what was this model trained on?
 - Would this be safe to use in a hospital? Why or why not?
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Part 2: Domain Mismatch

Now switch to a model that was fine-tuned for triage-style classification:

```

from transformers import AutoModelForSequenceClassification, AutoTokenizer, pipeline

model_id = "VolodymyrPugachov/BioClinicalBERT-Triage"
tokenizer = AutoTokenizer.from_pretrained(model_id)
model = AutoModelForSequenceClassification.from_pretrained(model_id)

classifier = pipeline(
    "text-classification",
    model=model,
    tokenizer=tokenizer,
    top_k=None
)

```

Discussion Prompts

Run the same clinical notes from Part 1 again. Then answer:

- How do the predictions change relative to Part 1?
 - Which label set is more meaningful for a healthcare task?
 - How was this model trained and what does it mean?
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Part 3: Clinical Summarization

Your Task

Use a summarization pipeline on a messy clinical note. Then inspect the output critically rather than assuming it is correct.

```

summarizer = pipeline("summarization")

```

```
note = ""
ED triage: 67-year-old male with HTN and type 2 diabetes presents with
chest pressure radiating to left arm for 2 hours after mowing lawn.
Reports nausea and diaphoresis. Home meds listed as lisinopril and metformin.
Allergies: NKDA.
```

Nursing addendum 15 min later: patient states he is 64, not 67. Says pain started yesterday, not today, and now denies radiation, nausea, or sweating. States he stopped taking all meds 3 months ago. Allergy listed as penicillin causes rash.

```
Resident note: wife says he nearly passed out in driveway and was clutching
his chest. Patient denies diabetes but prior chart shows A1c 9.1 last month.
Initial EKG documented as ST elevation in II, III, aVF; repeat note says
"no acute ST changes." Troponin pending.
""
```

```
print(summarizer(note, max_length=40))
```

Critical Angle

After you run the summary, identify at least one example of each of the following if present:

- Hallucinated details
 - Omitted critical facts
 - Overconfidence
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Part 4: “Break the Model” Challenge

Work with your group. Your goal is to design at least one input that makes one of the models behave badly.

You must:

- Create adversarial or ambiguous clinical inputs.
- Try to make the model fail.

Examples:

- Contradictory symptoms
- Negation (“no chest pain but...”)
- Rare diseases